

Sarcoma Subtype Reference

A Clinical Quick-Reference for Patients and Caregivers

Key diagnostic features, molecular markers, treatment approaches, and resources for the most common bone and soft tissue sarcoma subtypes.

Sarcoma Hub | Clinical Reference Series | Document 2 of 6
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How to Use This Reference

This document summarises key clinical facts for 14 sarcoma subtypes. It is intended as a starting point for patients seeking to understand their specific diagnosis, and for caregivers and advocates supporting them. Clinical management should always be guided by an expert multidisciplinary team.

Part A — Bone Sarcomas

Subtype	Key Features	Common Sites	Peak Age	Molecular	Treatment	Survival
Osteosarcoma	Bone-forming malignancy; most common primary bone cancer	Distal femur, proximal tibia, proximal humerus	Peak 10-20 yrs; second peak >65 yrs	None specific; complex karyotype	MAP (methotrexate, doxorubicin, cisplatin); limb-salvage surgery	~70% localised; ~20% metastatic
Ewing's Sarcoma	Small round blue cell tumour; EWSR1-FLI1 fusion in ~85%	Pelvis, femur, chest wall; any bone	5-25 yrs; rare in adults >40	EWSR1-FLI1 (85%), EWSR1-ERG (10%)	VAC/IE chemotherapy + surgery +/- radiotherapy	~70% localised; ~30% metastatic
Chondrosarcoma	Cartilage-producing; graded 1-3; chemo/RT resistant	Pelvis, femur, ribs, humerus	40-75 yrs; grade-dependent	IDH1/IDH2 (central type)	Wide surgical excision; no standard systemic therapy for grades 1-2	Grade 1: >90%; Grade 3: ~50%
Chordoma	Notochord remnant; locally destructive; brachyury positive	Skull base (35%), sacrum (50%), mobile spine (15%)	40-70 yrs	Brachyury (T gene) overexpression	En-bloc resection; proton beam RT preferred; targeted agents in trials	~68% 5-yr; high local recurrence
Giant Cell Tumour	Osteoclast-rich; locally aggressive; usually benign	Distal femur, proximal tibia, distal radius	20-40 yrs	H3.3 G34W mutation (>90%)	Curettage + local adjuvant; denosumab for unresectable/recurrent	>90% local control with expert surgery

Part B — Soft Tissue Sarcomas

Subtype	Key Features	Common Sites	Peak Age	Molecular	Treatment	Survival
GIST	Interstitial cells of Cajal; KIT or PDGFRA driven	Stomach (60%), small intestine (30%), other GI	Median 60 yrs	KIT exon 11 (65%), KIT exon 9 (10%), PDGFRA (15%)	Imatinib 1st line; sunitinib 2nd; regorafenib 3rd; avapritinib for PDGFRA D842V	~90% localised; ~50% advanced
Liposarcoma	Adipocytic; 4 subtypes with distinct biology and prognosis	Retroperitoneum, thigh	50-65 yrs	WD/DD: MDM2 amp (12q13-15); Myxoid: FUS-DDIT3 or EWSR1-DDIT3	Surgery; trabectedin for myxoid/round cell; CDK4 inhibitors in trials	WD: >90%; Myxoid: 70-80%; PI: ~50%
Leiomyosarcoma	Smooth muscle; uterine vs. non-uterine differ significantly	Uterus, retroperitoneum, vascular	50-70 yrs	Complex; RB1 loss common; no specific fusion	Doxorubicin +/- ifosfamide; gemcitabine-docetaxel 2nd line; trabectedin	~50-60% localised; ~25% metastatic
Synovial Sarcoma	SS18-SSX fusion; young adults; biphasic/monophasic	Extremities (periarticular), head/neck	15-40 yrs	SS18-SSX1, SS18-SSX2, or SS18-SSX4 fusion	Ifosfamide-based chemo; syrosingopine + metformin in trials	~70% localised; ~50% metastatic
Rhabdomyosarcoma	Skeletal muscle; 3 subtypes; most common STS in children	Head/neck, GU tract, extremities	Children; pleomorphic type in adults	Alveolar: PAX3/7-FOXO1; Embryonal: complex gains/losses	VAC chemotherapy; surgery; radiotherapy	~80% embryonal localised; ~65% alveolar localised
Angiosarcoma	Vascular endothelium; aggressive; post-radiation variant	Skin/scalp, liver, breast, soft tissue	60-70 yrs; de novo vs. secondary	MYC amp in radiation-associated; KDR mutations	Taxane-based therapy; sorafenib; surgery where feasible	~40% 5-yr; high recurrence and metastatic rate
UPS / MFH	Diagnosis of exclusion; no specific differentiation	Extremities, retroperitoneum	50-70 yrs	Complex; CDKN2A loss common	Doxorubicin-based; ifosfamide; surgical resection	~65% localised; ~20% metastatic
MPNST	Peripheral nerve sheath; 50% in NF1 patients	Extremities, trunk, head/neck	20-50 yrs	NF1 loss, PRC2 (EED/SUZ12) loss nearly universal	Wide resection; doxorubicin/ifosfamide; EZH2 inhibitors in trials	~50% NF1-associated; better in sporadic

This reference card provides a clinical overview only. Individual patient management must be guided by specialist multidisciplinary teams with access to full clinical and pathological information. Survival figures are approximate population-level estimates.